A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.12: IP Addressing (IPv4, IPv6, Subnetting, Public/Private, Static/Dynamic)

Total Questions: 17

Mark Schemes begin on Page 10

Question 1 | Source: 9618_w25_qp_11 (Q7.a)

7 Web browsers use Internet Protocol (IP) addresses.

(a) Complete the table by describing the following types of IP address.

Type of IP address	Description
Static	....
Public	....

[2]

Question 2 | Source: 9618_w25_qp_11 (Q7.b)

7 Web browsers use Internet Protocol (IP) addresses.

(b) Consider the following IP address:

256.0.0.A

Circle whether this IP address is IPv4, IPv6 or an invalid IP address.

IPv4 IPv6 Invalid

Justify your choice.

....

....

....

.... [2]

Question 3 | Source: 9618_s25_qp_12 (Q6.c)

6 A company has multiple sites in different cities.
The company has drivers who deliver products to customers. Each driver can connect to the company's WAN (Wide Area Network) whilst out of the office using a smartphone.

(c) Devices that connect to the WAN have an IP address.

Complete the following description of IP addresses by writing the missing words.

IPv4 is displayed as four groups of 8-bit numbers separated by ....

Each IPv4 address is 32 bits.

IPv6 is made of eight groups of 4 .... numbers separated by colons.

Multiple consecutive groups of .... can be replaced with a double colon.

Each IPv6 address is .... bits.

A .... IP address can change each time the computer connects to a network.

A .... IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.

[6]

Question 4 | Source: 9618_w24_qp_13 (Q9.d.i)

9 A computer is connected to a Local Area Network (LAN) that connects to a Wide Area Network (WAN).

(d) The LAN has a router. The router has a public IP address and a private IP address.

(i) State the purpose of a public IP address and a private IP address.

Public IP address ....

....

Private IP address ....

....

[2]

Question 5 | Source: 9618_w24_qp_13 (Q9.d.ii)

9 A computer is connected to a Local Area Network (LAN) that connects to a Wide Area Network (WAN).

(d) The LAN has a router. The router has a public IP address and a private IP address.

(ii) The LAN uses subnetting.

Describe the purpose of subnetting in a network.

....

....

....

.... [2]

Question 6 | Source: 9618_s24_qp_11 (Q8.b.i)

8 A business is creating a local area network (LAN) in its office.

(b) The LAN will connect to the internet through a router. The router has a public IPv6 address.

(i) State why the router has a public IP address.

....

.... [1]

Question 7 | Source: 9618_s24_qp_11 (Q8.b.ii)

8 A business is creating a local area network (LAN) in its office.

(b) The LAN will connect to the internet through a router. The router has a public IPv6 address.

(ii) One difference between an IPv4 and IPv6 address is that the numbers in an IPv4 address are separated by full stops and in an IPv6 address they are separated by colons.

Identify **two other** differences between an IPv4 and IPv6 address.

1 ....

....

....

2 ....

....

....

[2]

Question 8 | Source: 9618_s24_qp_13 (Q5.d)

5 A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

(d) The devices in the office have static private IP addresses.

State what is meant by a **static private IP address**.

....

....

.... [1]

Question 9 | Source: 9618_w23_qp_12 (Q7.a)

7 A Local Area Network (LAN) contains four devices:

- a router
- two laptop computers
- a server.

(a) The server has the IP address 192.168.3.2

Explain why this is **not** an IPv6 address.

....

....

....

.... [2]

Question 10 | Source: 9618_w23_qp_12 (Q7.b.iii)

7 A Local Area Network (LAN) contains four devices:

- a router
- two laptop computers
- a server.

(iii) Subnetting can be used to separate a network into logical segments.

Describe **two other** reasons why subnetting is used in a network.

1 ....

....

....

....

2 ....

....

....

....
[4]

Question 11 | Source: 9618_w23_qp_12 (Q7.d)

7 A Local Area Network (LAN) contains four devices:

- a router
- two laptop computers
- a server.

(d) The following incomplete table contains types of IP addresses and their descriptions.

Complete the table by writing the missing types of IP addresses and the missing descriptions.

Type of IP address	**Description**
....	an IP address that is assigned to a device to allow direct access on the internet
static IP address	....
....	an IP address used for internal LAN communication only
dynamic IP address	....

[4]

Question 12 | Source: 9618_s23_qp_11 (Q4.d.i)

4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

(d) The CCTV cameras are connected to a network and transfer their data wirelessly to the central computer.

(i) Each device on the network has an IP address.

Complete the description of IP addresses.

An IPv4 address contains .... groups of digits. Each group is represented in .... bits and the groups are separated by full stops.

An IPv6 address contains .... groups of digits. Each group is represented in .... bits. Multiple groups that only contain zeros can be replaced with a ....

[5]

Question 13 | Source: 9618_s23_qp_11 (Q4.d.ii)

4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

(d) The CCTV cameras are connected to a network and transfer their data wirelessly to the central computer.

(ii) The network makes use of subnetting.

Describe **two** benefits of subnetting a network.

1 ....

....

....

....

2 ....

....

....

....

[4]

Question 14 | Source: 9618_s23_qp_13 (Q2.e)

2 A university has two sites. Each site has several computer rooms. The computers are all connected as a WAN (wide area network).

(e) One site has split the network into several subnetworks.

An IP address in a subnetwork is divided into two parts.

Identify **and** describe the **two** parts of an IP address in a subnetwork.

....

....

....

....

....

.... [3]

Question 15 | Source: 9618_w22_qp_13 (Q7.c.i)

(c) Local Area Networks (LANs) can be made up of several subnetworks.

(i) Give **two** benefits of dividing a network into subnetworks by subnetting the LAN.

1 ....

....

2 ....

....
[2]

Question 16 | Source: 9618_w22_qp_13 (Q7.c.ii)

(c) Local Area Networks (LANs) can be made up of several subnetworks.

(ii) A subnet mask is used when subnetting a LAN.

Two devices on the LAN are located in different subnetworks.

The IP addresses and corresponding subnet masks are shown:

Device IP address	Subnet mask
`10.10.12.1`	`255.0.0.0`
`192.168.12.4`	`255.255.255.0`

Identify the following network ID and host ID.

The network ID for the device with the IP address `10.10.12.1`

....

The host ID for the device with the IP address `192.168.12.4`

....
[2]

Question 17 | Source: 9618_s21_qp_12 (Q5.d)

5 Seth uses a computer for work.

(d) Draw **one** line from each term to its **most appropriate** description.

Term	Description
	It is only visible to devices within the Local Area Network (LAN)
Public IP address	It increments by 1 each time the device connects to the internet
Private IP address	A new one is reallocated each time a device connects to the internet
Dynamic IP address	It can only be allocated to a router
Static IP address	It is visible to any device on the internet
	It does not change each time a device connects to the internet

[4]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q7.a)

7(a)	**1 mark** for each correct description, **max 2** marks <table><tr><th>Type of IP address</th><th>Description</th></tr><tr><td>Static</td><td>IP address that does not change each time a device **connects to the network**</td></tr><tr><td>Public</td><td>IP address assigned to a device on a network to allow the device to be visible on the internet</td></tr></table>	**2**

Mark Scheme for Question 2 | Source: 9618_w25_ms_11 (Q7.b)

7(b)	**1 mark** per bullet point for justification, **max 2** marks Choice: Invalid (no mark for the choice) Justification: • It is not IPv4 because the maximum denary value should be 255, this uses 256 • It is not IPv6 because it uses a full stop to separate sections and not colon // It is not IPv6 because there are not enough groups	**2**

Mark Scheme for Question 3 | Source: 9618_s25_ms_12 (Q6.c)

6(c)	**1 mark** for each correct word • **full stops** • **hexadecimal** • **zeros** • **128** • **dynamic** • **private** IPv4 is made of four groups of 8-bit numbers separated by **full stops**. Each IPv4 address is 32 bits. IPv6 is made of eight groups of 4 **hexadecimal** numbers separated by colons. Multiple consecutive groups of **zeros** can be replaced with a double colon. Each IPv6 address is **128** bits. A **dynamic** IP address can change each time the computer connects to a network. A **private** IP address can only be accessed by other devices in the same LAN and is assigned by the router within the LAN.	**6**

Mark Scheme for Question 4 | Source: 9618_w24_ms_13 (Q9.d.i)

9(d)(i)	**1 mark** for public IP address and **1 mark** for private IP address: Public IP address: • So that the router is visible to the Internet/external network/WAN Private IP address: • So that the router is identified to computers within the LAN	**2**

Mark Scheme for Question 5 | Source: 9618_w24_ms_13 (Q9.d.ii)

9(d)(ii)	**1 mark** for each bullet point (**max 2**) • It allows the network to be divided into smaller networks • … which reduces traffic in some parts of the network // reduces congestion • … because traffic only travels through the parts necessary • … which hides the complexity of network • … and allows for easier maintenance of the network	**2**

Mark Scheme for Question 6 | Source: 9618_s24_ms_11 (Q8.b.i)

8(b)(i)	**1 mark** for: to **be visible** to and accessible by other devices on the internet	**1**

Mark Scheme for Question 7 | Source: 9618_s24_ms_11 (Q8.b.ii)

8(b)(ii)	**1 mark** each: • IPv4 has 4 groups of digits whilst IPv6 has 8 groups • IPv4 is usually represented in **denary** whilst IPv6 is usually represented in **hexadecimal** • IPv4 groups are between 0 and 255 whilst IPv6 is between 0 and FFFF • IPv4 is **32** bits whilst IPv6 is **128 bits**	**2**

Mark Scheme for Question 8 | Source: 9618_s24_ms_13 (Q5.d)

5(d)	**1 mark** for: Static means the IP for that device does not change **and** Private means it can only be accessed/seen/used within the LAN	**1**

Mark Scheme for Question 9 | Source: 9618_w23_ms_12 (Q7.a)

7(a)	**1 mark** for each bullet point **(max 2)** • Only has four groups of digits // IPv6 has eight groups • Uses dotted notation instead of colons • Because it is a 32 bit / 4 byte address // IPv6 is 128 bits / 16 bytes	**2**

Mark Scheme for Question 10 | Source: 9618_w23_ms_12 (Q7.b.iii)

7(b)(iii)	**1 mark** for each use (**max 2**) and **1 mark** for corresponding expansion (**max 2**) • To improve the security of the LAN • … so that devices do not receive unintended data • … so that a compromised device does not expose the whole network • ... so not all devices can access all segments • To make the network management easier • … because faults can be isolated more efficiently • ... by appropriate example • To make the network easier to expand // For better control of network growth • … by allowing for greater range of IP addresses to be available • To improve network performance • To reduce network congestion • … by localising network communications // by dividing data between segments • … so that devices are not flooded with data • … because data sent between devices on the same subnet stays within the subnet	**4**

Mark Scheme for Question 11 | Source: 9618_w23_ms_12 (Q7.d)

7(d)	**1 mark** for each highlighted area **Type of IP address** / **Description** public IP address — an IP address that is assigned to a device to allow direct access on the internet static IP address — an IP address that is fixed / doesn't change each time a device re-joins a LAN / network private IP address — an IP address used for internal LAN communication only dynamic IP address — an IP address that may be refreshed / changed each time a **device re-joins** a LAN / network	**4**

Type of IP address	Description
public IP address	an IP address that is assigned to a device to allow direct access on the internet
static IP address	an IP address that is fixed / doesn't change each time a device re-joins a LAN / network
private IP address	an IP address used for internal LAN communication only
dynamic IP address	an IP address that may be refreshed / changed each time a **device re-joins** a LAN / network

Mark Scheme for Question 12 | Source: 9618_s23_ms_11 (Q4.d.i)

4(d)(i)	**1 mark** for each term An IPv4 address contains **4** groups of digits. Each group is represented in **8** bits and the groups are separated by full stops. An IPv6 address contains **8** groups of digits. Each group is represented in **16** bits. Multiple groups that only contain zeros can be replaced with a **:: // double colon**.	**5**

Mark Scheme for Question 13 | Source: 9618_s23_ms_11 (Q4.d.ii)

4(d)(ii)	**1 mark** for identification, **1mark** for expansion e.g. • Reduce amount of traffic in a network // improve network speed • Data stays in its subnet so it does not travel as far • Improves network security • .. so that not all devices can access all areas of the network • Allows for easier maintenance • … because only one subnetwork may need taking down/changing while the rest of the network can continue	**4**

Mark Scheme for Question 14 | Source: 9618_s23_ms_13 (Q2.e)

2(e)	**1 mark** for identification • IP address is made up of a <u>network ID</u> and a <u>host ID</u> **1 mark** each to **max 2** for description • Each device in a subnetwork has the same network ID // Each subnetwork has a different network ID • Every device **in each subnetwork** has a different host ID but the same network ID // the host ID uniquely identifies the device **within the subnetwork**	**3**

Mark Scheme for Question 15 | Source: 9618_w22_ms_13 (Q7.c.i)

7(c)(i)	**1 mark** for each bullet point (**max 2**) Examples: • improves security • reduces congestion • allows extension of the network / devices attached • aids day-to-day management • improves performance	**2**

Mark Scheme for Question 16 | Source: 9618_w22_ms_13 (Q7.c.ii)

7(c)(ii)	**1 mark** for each correct answer: • network ID = 10 • host ID = 4	**2**

Mark Scheme for Question 17 | Source: 9618_s21_ms_12 (Q5.d)

5(d)	**1 mark** for each correct line	**4**